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Chord keyboards

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Unlike the standard typewriter keyboard, keying on a chord keyboard is carried out by simultaneous patterned pressing of one or more keys. This results in fewer keys being needed on a chord keyboard when compared with a sequential keyboard, where keys are pressed one at a time. For example, five keys allow a total of 31 ($2^5 - 1$) different chord combinations to be generated.

The feasibility of using a chord keyboard as a data entry input device was first seriously investigated in the mid-1950s by the Canadian Post Office. The trend towards research into chord keying reached a peak around 1960 when International Business Machines (IBM) studied two chord keyboards to rival the typewriter. It was not until the 1970s that chord keyboards became commercially available and within the last decade three chord keying devices have been marketed. The emphasis on the development of these recent keyboards has moved from task specific to more general purpose applications.

This paper (based upon Martin (1980) the author's maiden name) reviews the chord keyboards which have been developed since the 1940s with special emphasis on the mail sorting application, and draws some conclusions concerning future developments.

Keywords: Keyboards, typewriter, mail sorting

Introduction

At present the standard QWERTY keyboard provides the most effective man-computer communication input device. It has been suggested that the chord or 'multipress' keyboard may be a useful alternative for some systems. As the name suggests, one or more fingers are needed to key a single character. In its simplest form, a chord keyboard consists of five keys, one for each finger and the thumb. Such a device enables 31 different characters to be transmitted. If a second thumb key is added, this increases the number of alternative chord combinations to 47. It is possible with a two-handed chord keyboard to make available 1023 different combinations.

In the formative years of chord keying, development was mainly concerned with systems for the sorting of mail. The British, Canadian and American postal services experimented extensively with the use of chord keyboards. It is only within the last decade that the chord keyboard has been considered for wider use, even perhaps as a substitute for the pen and typewriter (Shackel, 1976).

History of chord keying

Early attempts at chord keyboard design

A patent for a mechanical typewriter, which operated on the principle of a chord keyboard, was requested by Achille Colombo in 1942. This worked on the principle of one left-hand and one-right hand key being pressed simultaneously to type a character. However, no further details of this device have been found in the open literature and it is thought that the machine was never marketed (Conrad and Longman, 1965).

One of the first chord keyboards to be reported was by Dvorak (1950). Unfortunately no details of this one-handed keyboard exist, and no information is available on its design or application (Seibel, 1972).

Levy (1955) was one of the first to describe a 10-key 'binary' keyboard for use at the Toronto Post Office. This keyboard consisted of two 5-key keyboards and by pressing the correct combination of keys, the addresses could be condensed into a series of characters. The use of chord keying, combined with a method of encoding addresses, resulted in a higher speed of sorting than manually. The contribution of the chord keyboard to this is not apparent.

Five years later, Conrad (1960) reported a letter sorting machine involving simultaneous depression of two keys (one by each hand) in order to sort letters into one of 144 possible destinations. Each hand had one of 12 keys to select and it was found that sorting rates improved from approximately 35 sorts/min for the manual system to about 60 sorts/min using the keyboard over a practice period of 39 weeks.

Cornog, Hockman and Craig (1963) reported a study of their 'double-binary' chord keyboard, which was used in America for the sorting of mail. Keying on this keyboard

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produced a binary code, and it has been suggested that the keyboard was inappropriately named 'double-binary'. The keyboard training console consisted of four banks of six keys, an arm-rest bar and a panel of lights. After 36 weeks of practice, the experimental results suggested that the keyboard was a satisfactory proposition for mail sorting.

Cornog and Craig (1965) described several chord keyboards for use in mail sorting. These included the Levy keyboard, the Burroughs keyboard and the so-called FMC keyboard. The Burroughs keyboard consisted of 10 piano-like keys, which enabled mail to be sorted to 3000 destinations. Chord combinations were assigned to numbers, which were used to represent the 3000 destinations. The average output was 45 chords/min with a 2% error rate. In comparison the FMC keyboard had two banks of 12 keys, and it was found that two-handed operation of this keyboard was about 25% faster than one-handed. However, due to variations in the training periods for the three keyboards, it was not possible to draw any firm conclusions from the results.

There have been other chord keyboard developments besides those involved with mail sorting. Klemmer (1958) of IBM studied a 10-key chord keyboard with five keys arranged in a semi-circle for each hand. He allotted the most frequently occurring letters in the English language to single keys, while other characters were represented by depressing two of the 10 keys. Klemmer trained two subjects on his keyboard, one of whom was a skilled typist. He found that after 40 h of training, the typist had reached a speed of 47 words/min with a 0·3% error rate, compared with 29 words/min and 0·7% errors for the naive typist. It was concluded that the learning curves for the acquisition of chord keying were comparable with those expected on a conventional typewriter.

In 1959, IBM carried out an evaluation of an 8-key word writing typewriter. This chord keyboard consisted of two arcs of four keys, and the thumbs were not used. Four subjects learned the 137 chord patterns in less than 30 h. Lockhead and Klemmer (1959) concluded that "the ability of people to learn and their willingness to use multiple key patterns is encouraging".

One of the most comprehensive studies of chord keyboard design was carried out by Hillix and Coburn (1961) at the Navy Electronics Laboratory. They compared seven different designs of keyset and used time and error measures combined in a single performance index to indicate the number of bits of information transmitted per second. They suggested that the ideal keyset should have the following optional features incorporated into the design: a touch operation facility, a hand-configured layout, provision for two-handed operation with maximum alteration facility, provision for individual character error correction and an easy-to-learn operating procedure.

After an extensive statistical analysis of letter, digram and trigram frequencies, Ayres (1965) patented his word writing machine. This device was designed for chord keying and the keys were so arranged that they could be operated simultaneously. Typing worked on the principle that letters would be selected from each vertical column of five characters from left to right across the keyboard. By selecting these initial consonants, intermediate vowels and final consonants, theoretically whole words could be printed with one stroke of the keys (see Fig. 1).

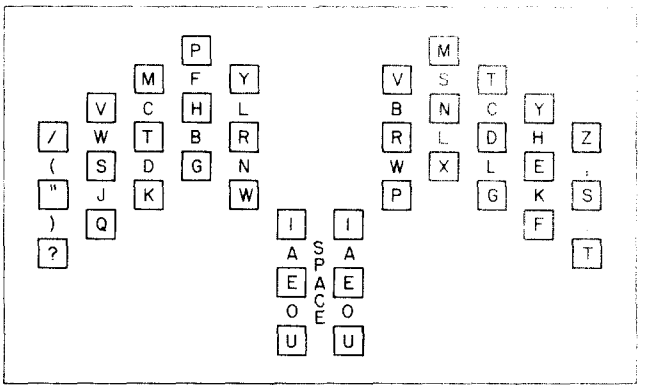

Fig. 1 Ayres' word writing machine (1965)

However, as with most chord keyboard developments, no evaluative work was carried out and apart from US Patent No 3,225,883 no other information concerning this device has been found in the open literature.

Englebart in April 1966 developed a 5-key keyboard for one-handed entry of alphabetic data (Barmack and Sinaiko, 1966). Little information is available about this keyboard, although it is known that it took approximately 10 h for Englebart to reach a speed of 10 words/min. After more training, he was able to type at a speed of 35 words/min with his right hand, and 25 words/min with his left hand.

Experimental comparisons of sequential and chord keyboards

There have been several keyboard comparisons conducted (for example, Levy (1955) compared the speed of typing of three chord keyboarders with a skilled typist, and Klemmer (1958) studied the learning curves of an unskilled and a skilled typist using a 10-key chord keyboard), but the most comprehensive studies were two experiments carried out in 1965.

Conrad and Longman (1965) compared the performance of two groups of postmen on a chord and a typewriter keyboard. The chord keyboard had 10 'home' keys, which were depressed in pairs using the left and right hands. The time allowed to depress the two keys was 50 ms, and if more keys were depressed or the 50 ms ran out, an error was registered. Both these situations did not arise with the standard keyboard.

Postmen in the 30–40 years age group participated in the experiment and 24 individuals were allocated at random to the typewriter group and 22 to the chord group. No subjects had prior typing experience and they trained for 3½ h daily, five days a week for seven weeks. This seven week course took place on four occasions making the total length of the experiment about 10 months.

Conrad and Longman found that the group using the chord keyboard became 'operational' about two weeks earlier than the standard typewriter group, that is, in two weeks instead of four and after that their improvement rates could be regarded as parallel.

While Conrad and Longman were experimenting at the British Post Office, Bowen and Guinness (1965) were attempting to design the optimal keyboard for the sorting of mail in the United States. They experimented with two chord keyboards having 12 and 24 keys each, and a standard typewriter keyboard. A series of pilot studies using small numbers of subjects (three to six) were conducted and it

was concluded that the chord keyboard was equal, if not superior, to the sequential keyboard for mail sorting.

Chord keyboard developments within the last decade

Between 1950 and 1970, the majority of chord keyboards developed were for specific tasks such as mail sorting. Only IBM explored the possibility of using a chord keyboard instead of a sequential layout, and hence they called their developments the '10-key typewriter' and the '8-keyword writing typewriter'. Throughout the last decade, rapid electronic advances have been made and a trend towards personal computing has begun to develop. Hence, it is not surprising that the chord keyboard has been developed with the view to being a personal connection to a computer for general use.

One of the first researchers to recognise the potential of the chord keyboard was Stewart (1973). He patented the ANTEL chord keyboard. This 'calculator-like' device was basically a 12-key keyboard, which was operated as a sequential keyboard, when one of the 12 keys was depressed, and as a chord keyboard, when an appropriate chord or group of keys was depressed simultaneously (see Fig. 2). The letter A is obtained by pressing the key labelled A, while B is obtained by pressing the keys A and C. It is possible to press up to four keys with one finger, due to the compactness of the keyboard. The letters G and I would be obtained in this way.

Within the last three years, three chord keyboards have been developed. In January 1978, details of a hemispherical chord keyboard called the 'Writehander' were published by Owen (1978). This one-handed keyboard has four finger keys and eight keys for the thumb. By using various combinations of fingers and thumb, the Writehander allows the entire 128 characters of the ASCII code to be generated. Like ANTEL, the Writehander was intended for interfacing with "computers, terminals, displays and other ASCII coded devices". No details of the performance levels attainable with these devices are available in the open literature.

IBM has also recently developed a chord keyboard (Rochester, Bequaert and Sharp, 1978). This miniature keyboard is operated with one hand and is suitable for typing large amounts of data at high speed. Bequaert and

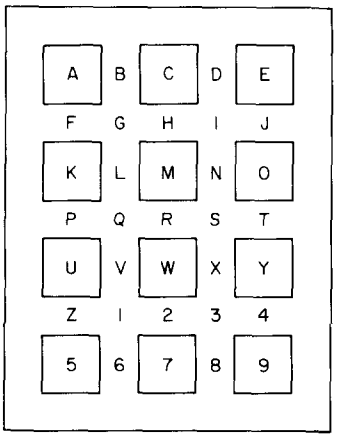

Fig. 2 Diagrammatic representation of the ANTEL keyboard.
NOTE: Zero is represented by the letter 0. There are 35 clearly labelled functions.

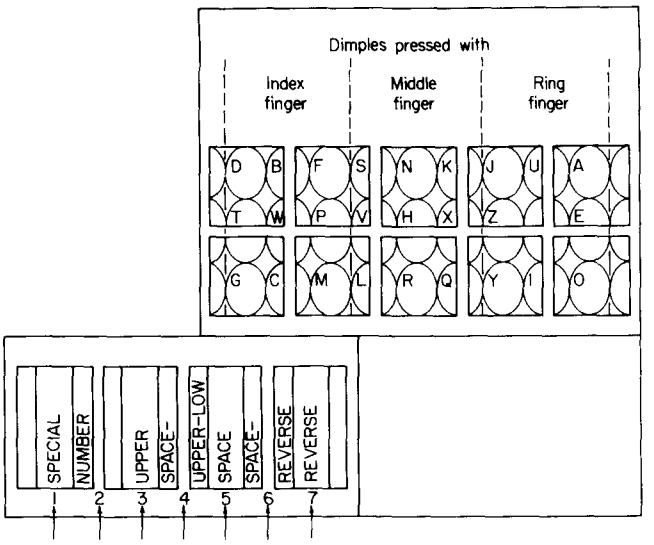

Fig. 3 The IBM chord keyboard

Rochester (1977) described an automated system for the teaching of touchtyping using this keyboard. A right handed version is shown in Fig. 3. It has a five by two array of square keys operated by the fingers and a row of four rectangular keys operated by the thumb. The finger keys have rounded depressions called 'dimples'. The IBM chord keyboard works on similar principles to the ANTEL keyboard. If the B dimple is pressed, two keys go down and a B is typed, while depressing the W dimple results in four keys being pressed and a W being typed. The keyboard consists of 27 dimples, which allows the full alphabet to be typed.

The IBM keyboard differs from the Writehander in that a three finger chord would produce a string of three letters. The dimples have been arranged so that the fingers can produce common trigrams such as 'the' in one key-pressing. The little finger is redundant since it does not function well independently from the ring finger.

The thumb has a separate keyboard consisting of seven troughs, (see numbered arrows) and as 'no response' from the thumb is a valid action, the thumb has eight possible positions. By pressing a trough it is possible to depress either one thumb key or two adjacent thumb keys as part of a chord. The following example of a typical sentence demonstrates the chords to be formed.

/I/n/ thi/s se/g/me/nt/ of/ te/xt/ the/ cho/rd/
bo/und/ar/i/es/ ha/ve/ be/en/ s/ho/wn/ by/ li/ne/s.

Rochester, Bequaert and Sharp (1978) have shown that a number of students have learned to type on this chord keyboard at a gross typing speed of over 40 words/min.

The most recent chord keyboard to be developed is the Microwriter (Endfield, 1978). The Microwriter (see Fig. 4) is a portable, battery operated, chord keyboard about the size of a medium-sized electronic calculator. There are five basic keys (one for each finger and thumb), plus a sixth key which controls functions, enters figures, punctuation marks and editing control, and a key for entering capital letters. There are 16 editing functions including back-space, insert, delete, etc. The Microwriter was designed to be a "personal, portable writing instrument, with full text editing facilities" (Mussin, 1980).

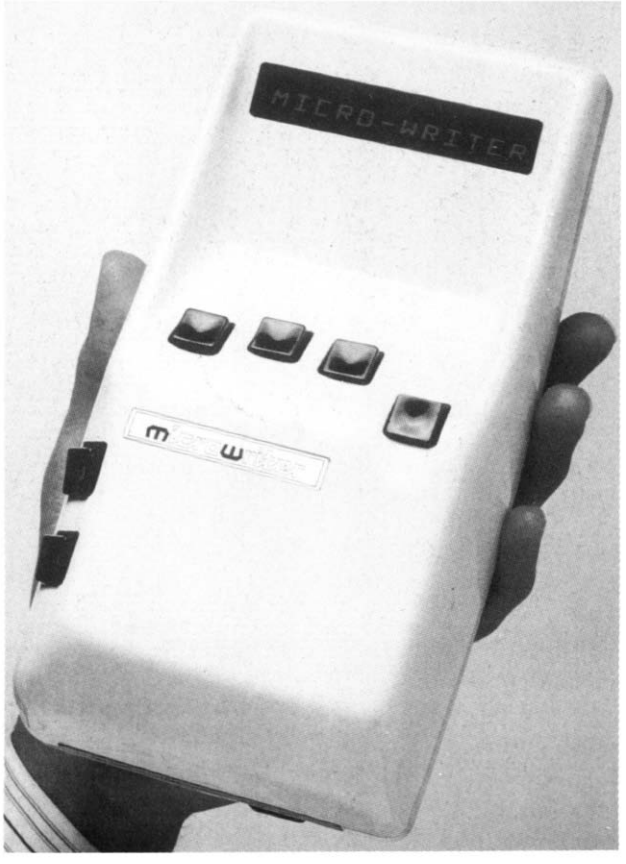

Fig. 4 The Microwriter

Characters are keyed in one at a time, and in order to do this the various chords for each character have to be learned. Endfield has chosen combinations of keys that he claims are easy to remember because the shape of the letters are made out of the five keys. Like the Writehander, the more common letters are allocated easier chords, but some mnemonics, for example, F, M and Q are not particularly representative of their letters.

The Microwriter has a single line 16-segment LED display of up to 12 characters. The text is recorded in the machine's 8K byte RAM memory, which is able to store up to 8000 characters. Extra storage can be provided on cassettes. The Microwriter does not itself produce words on paper, but plugs into an electronic typewriter, a television set, or a computer system. Endfield, the inventor, stated that learning to use this chord keyboard takes about 30 min. No published, controlled evaluation of this keyboard is available, although Endfield claimed that after using his device for a period of seven months, he was keying at a speed of about 40 words/min. At Newcastle-upon-Tyne Polytechnic, six lecturers learned to operate the Microwriter (Wheeler, 1980) over a four month period They prepared on average five pages of A4 text each week, using the Microwriter on two or three occasions. The conclusions from this study favoured the use of the Microwriter, although many of the advantages listed were concerned with word processing and did not arise as a direct result of using a chord keyboard.

Chord keyboards – an appraisal

It becomes apparent that some chord keyboards have been developed with no specific use in mind other than general purpose applications. The advertising attached to these keying devices is usually concerned with the fact that they are superior to the standard QWERTY keyboard (for example, "QWERTY is obsolete" to advertise the Writehander, "A Typewriter in your Pocket" for the Microwriter). In order to try and clarify this situation, some advantages of using a chord keying device are considered.

(a) The option of one or two handed use. Unlike a sequential keyboard, a chord keyboard can be operated by one hand. The advantages of a one-handed keyboard are:

- the device does not take up space immediately in front of the operator
- the user does not have to sustain an unnatural keying posture
- the keyboard can be moved freely over a wide area
- the other hand is free to perform a second task such as answering the 'phone, turning the pages in a book, etc.

(b) The comparative small size and compactness of a chord keyboard enhances its portability and potential usability in 'hostile' environments, for example, whilst travelling when writing or dictation would not be feasible.

(c) In terms of information theory, chord keyboard users enter more bits of data per second than sequential keyboard operators. However, this is misleading, because in real terms, less characters per minute are being emitted. An accurate picture of the potential of chord keyboards cannot be realised, because there are few skilled chord keyboarders on which to collect speed measurements. Although it is known that stenographers have one of the fastest modes of data entry, this is partly because they transcribe text into a form of shorthand.

(d) It has been suggested by Endfield (1978) that chord keyboards are easier to learn than sequential keyboards, because the fingers never have to move between the keys. However, there is no definitive research evidence to support or disprove this claim.

(e) Chord keyboards have a novelty appeal, which may result in sections of the population who would not normally learn to type being attracted to them, for example, executives and office staff. However, is the novelty of a chord keyboard sufficient basis to guarantee its commercial success? The overriding problem with devices which are purchased because of their 'fun to use' and novelty, is that with time these factors fade in significance.

Many of the advantages of chord keyboards stem from their size, and one-handed operation, which increases their flexibility. It would appear unlikely that ease of learning would be an aspect favouring a chord keyboard when compared with a sequential keyboard, since chord keyboards require initial guidance before they can be used, followed by intensive learning of the chord patterns.

There is a lack of application associated with the development of the recent chord keying devices, and this raises the issue of whether there is a need for chord keyboards other than those which are designed for specific tasks. At present, there seems little justification for

developing these 'general purpose' chord keyboards, since the sequential keyboards available fulfil the majority of the requirements of the general population. But it is inevitable that inventors and enterprising individuals will continue to develop chord keyboards, and it could be argued that although there appears to be no immediate need for a general purpose chord keyboard, the rapid development of personal computers might create the need.

The chord keyboard is still in an embryonic state compared to the century old sequential QWERTY keyboard; it merits further attention and research to prevent a repeat of the 'QWERTY episode', where one of the first keyboards to be developed becomes recognised as the standard. As long ago as 1963, Pollock and Gildner suggested the need for research into chord keying, before enthusiasm for a new and novel input technique might lead to improper incorporation into a computer system.

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